Topic #142 — Leave-One-Out Cross-Validation

Train on $N-1$ samples, test on 1, repeat N times. When is it appropriate? What does it cost?

1. Intuition

Any model has a temptation: *memorise* the training data rather than learn its underlying pattern. Evaluating such a model on the same data it trained on gives a falsely good score — this is overfitting. What we actually want is the **test error**: performance on data the model has never seen.

The standard fix is a held-out test set, but when N is small (40–80 points), reserving 20% for testing means training on far fewer samples, which itself introduces bias into the evaluation. You are stuck: use more data to train and your evaluation is unreliable; use more data to evaluate and your model is weaker.

The LOO idea. For each data point in turn: hide it, train on everything else, predict on the hidden point, record the error. Repeat for all N points, then average. No data is ever wasted — every point is test set exactly once and training set $N-1$ times. The result is *deterministic*: no random splits, so running it twice gives the same answer.

Because each training set has $N-1 \approx N$ points, each individual error is nearly unbiased for the true test error. Averaging N such estimates makes the final number stable.

2. Mathematical & Algorithmic Core

Require: Dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, algorithm $\mathcal{A}$, loss L

for $i = 1$ **to** N **do**
  $\hat{f}_{-i} \leftarrow \mathcal{A}(\mathcal{D} \setminus \{(\mathbf{x}_i, y_i)\})$ // *fit on $N-1$ points, excluding obs. i*
  $e_i \leftarrow L\Big(y_i,\ \hat{f}_{-i}(\mathbf{x}_i)\Big)$ // *error on the one held-out point*
end for
return $\mathrm{CV}_{(N)} = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} e_i$

The estimator. For squared loss: $\mathrm{CV}_{(N)} = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N}(y_i - \hat{y}_{-i})^2$, where $\hat{y}_{-i}$ is the prediction of the model trained *without* point i. Because (x_i, y_i) had no role in fitting, each term is approximately unbiased for the true test MSE.

Hat-matrix shortcut (linear models). Naively LOO requires N separate model fits — expensive. For ordinary least-squares, a single fit suffices (James et al. 2013, Eq. 5.2):

$$\mathrm{CV}_{(N)} \;=\; \frac{1}{N}\sum_{i=1}^{N}\left(\frac{y_i - \hat{y}_i}{1 - h_i}\right)^2$$

where $h_i = \mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i \in [1/N, 1]$ is the *leverage* of observation i, and $\hat{y}_i$ is the ordinary fitted value. A high-leverage point has outsized influence over its own fitted value; dividing by $(1 - h_i)$ inflates its residual by exactly the right amount to simulate leaving it out. **This reduces LOOCV for linear models to a single matrix computation.**

Numerical dry-run. Suppose $N = 4$ with leave-one-out squared errors $e_1 = 0.04$, $e_2 = 0.09$, $e_3 = 0.01$, $e_4 = 0.16$. Then $\mathrm{CV}_{(4)} = \frac{1}{4}(0.04 + 0.09 + 0.01 + 0.16) = 0.075$ — our unbiased estimate of generalisation error.

3. Visual Element

Figure 1. LOO on $N = 6$. • train; × test (held-out).

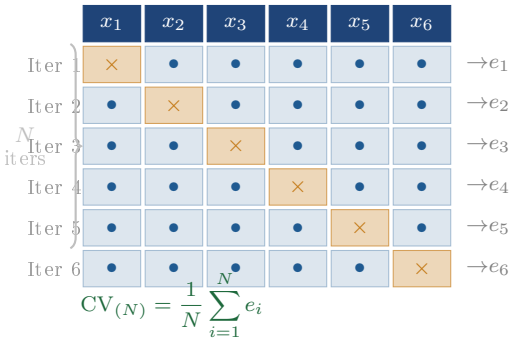

Table 1. Comparison of CV strategies.

Method	Train size	Bias	Var.	Cost
LOO (k=N)	$N-1$	Low	High	$N{\cdot}T$
k-fold (k=10)	$0.9N$	Low	Mod.	$k{\cdot}T$
Val. set	$0.5N$	High	High	T

Why is LOO variance high? All N training sets share $N-2$ identical points, so the N error terms are *highly correlated*. Averaging correlated estimates barely reduces variance — this is why k=10 fold often outperforms LOO despite slightly more bias.

4. When to Use LOO — and When Not To

Use LOO when: (1) N is small ($N < 100$) — every sample is precious, no data can be wasted on a fixed test split; (2) you need a *deterministic, reproducible* error estimate (scientific reporting, clinical data); (3) you are using a *linear model* — the hat-matrix shortcut makes it computationally free.

Avoid LOO when: (1) N is large: if N=10,000 and training takes 2 minutes, LOO costs ≈14 days of compute; (2) the model has high intrinsic variance (deep networks, gradient boosting) — the N error estimates will themselves be unstable; (3) a fast approximate estimate suffices — k=10 fold is $N/10$ times cheaper for comparable accuracy.

Cost in one line: LOO cost $= N \times T$ vs. k-fold cost $= k \times T$; for k=10, N=1000: LOO is **100× more expensive**.

5. Connections to Broader Course Concepts

Bias–Variance Tradeoff. Choosing a CV method is itself a bias-variance problem. LOO minimises bias in the *estimator* by maximising training-set size, but pays with high variance from correlated folds. k=10 fold sits at the practical sweet spot — slightly more bias, much lower variance — mirroring the same principle that justifies regularisation over interpolation in model fitting.

The Bootstrap. Both LOO and the bootstrap generate N estimates from a single dataset. The bootstrap resamples *with replacement* (some points appear multiple times, some not at all); LOO resamples *systematically without replacement*. Both address the same problem: estimating generalisation error without a dedicated test set.

Model Selection and AIC/BIC. Plotting LOO error as a function of model complexity (polynomial degree, regularisation strength, tree depth) gives a non-parametric selection curve. AIC and BIC solve the same task analytically under specific distributional assumptions; LOO works assumption-free, which is why it is preferred when those assumptions may fail.

References

[1] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An Introduction to Statistical Learning: with Applications in R*. Springer.
Chapter 5: Resampling Methods, §5.1.2, pp. 178–182. Free PDF: https://www.statlearning.com
Primary reference. Contains the LOO estimator derivation (Eq. 5.1), the hat-matrix shortcut that makes LOO free for linear models (Eq. 5.2), the iteration schematic (Fig. 5.3), and the empirical LOO-vs-kfold comparison on the Auto dataset (Fig. 5.4). All mathematics in Sections 2 and 3 above traces to this chapter.

[2] StatQuest with Josh Starmer. (2018). *Machine Learning Fundamentals: Cross Validation* [Video]. YouTube.
https://www.youtube.com/watch?v=fSytzGwwBVw (LOO segment: 3:30–5:00)
Best intuition resource. Explains LOO and k-fold visually with minimal assumed background. The framing in Section 1 — building the problem before naming the technique — follows Starmer's pedagogical approach. Recommended as a first watch before reading the textbook.

[3] Scikit-learn Development Team. (2024). *3.1. Cross-validation: evaluating estimator performance.* Scikit-learn User Guide.
https://scikit-learn.org/stable/modules/cross_validation.html
https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.LeaveOneOut.html

Formal definition and working implementation using `LeaveOneOut()` + `cross_val_score()`. Confirms that the "N model fits" description maps literally to code, and shows how LOO integrates with sklearn Pipelines to avoid data-leakage in preprocessing.

[4] Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning* (2nd ed.). Springer. Chapter 7, §7.10. Free PDF: https://hastie.su.domains/ElemStatLearn/
Supplementary for deeper theory. Specifically: the formal proof that averaging N near-identical LOO training sets inflates estimator variance relative to k-fold (the correlation argument referenced in the Table 1 discussion), and the warning about "wrong" vs "right" applications of cross-validation in feature selection.